Project Seminar WS2026/27

Design of a fuel cell monitoring board in KiCad and development of the operational board software

Lesson 3: Fundamentals PCB-Design and project introduction

M. Eng. Andreas Hein

E-Mail: andreas.hein@hs-wismar.de

FIW: Fachbereich Elektrotechnik und Informatik

HOCHSCHULE WISMAR
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Project Seminar – Report

- ***General structure (1):***
  - Page 1:
    → “Project Seminar WS2026/27”
    → Topic: “Design of a fuel cell monitoring board in KiCad and development of the operational board software“
    → Face-photo
    → Name
    → Matriculation number
  - Page 2: Table of content
    → Chapter 1: “PCB-manufacturing”
    → Chapter 2: “Assembly technologies and PCB-components”
    → Chapter 3: …
    → Chapter 4: …

Project Seminar – Report

- ***General structure (2):***
  - Page 2: Table of content
    - → Chapter 1: PCB-manufacturing
    - → Chapter 2: Assembly technologies and PCB-components
    - → Chapter 3: Basics in PCB-Design
    - → Chapter 4: PEM-Fuel-Cell
    - → … (put here all further chapters in)
    - → Exercise 1: Reference voltage source
    - → Exercise 2: Successive-approximation ADC (SAR ADC)
    - → … (put here all further exercises in)
    - → List of figures
    - → List of acronyms
    - → List of literature (used documents, books, papers, websites and other sources!)

Project Seminar – Report

- ***General structure (3):***
  - Every new chapter begins on a new page!
  - The solution to every exercise begins on a new page!
  - Font: Arial or Calibri
  - Minimum font size: 12
  - Every figure and every diagram and every equation and every table must be good readable!
  - Every equation/formula is assigned a sequential number!
    → y= a +b (Eq. 1.1) → *means equation 1 in chapter 1*
  - Every figure/diagram is assigned a sequential number!
    → Figure 1: *(Caption of the figure)*
  - Every table is assigned a sequential number!
  - → Table 1: *(Caption of the table)*

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- ***... very important! → restrictions!:***
  - **Don't use** AI-generated full sentences, phrases, text passages, images or diagrams!
  - **Do not use** any AI-generated formulas, equations, or source code!
  - **Do not copy** solutions or work from others!

Project Seminar 2026

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Project Seminar – Basics PCB Design

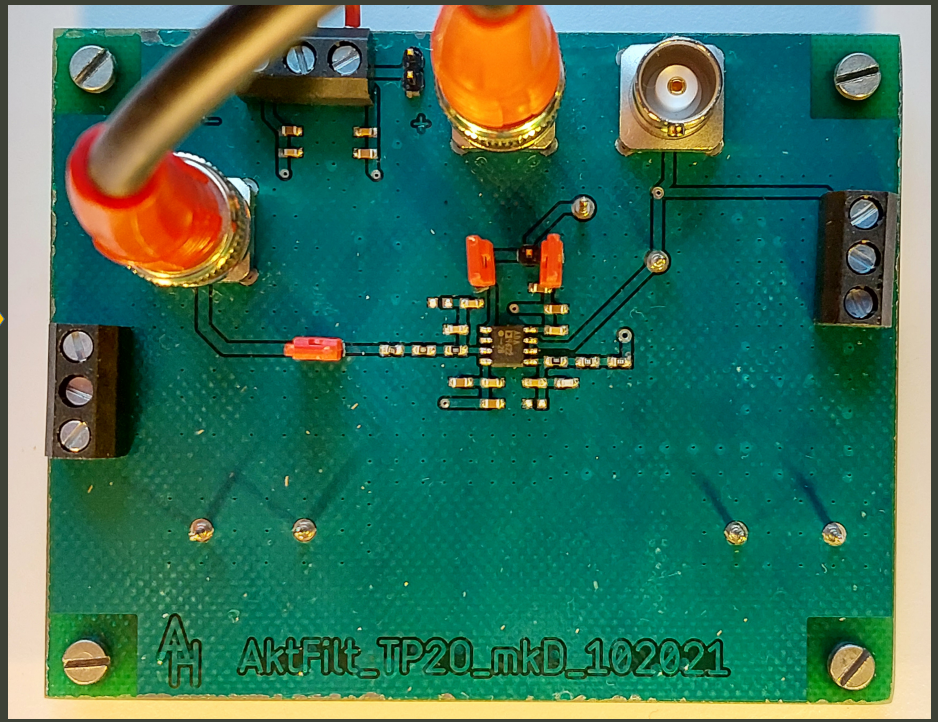
- ***Goal: design of Printed Circuit Board for an certain application***
  - What requirements must the electronic circuit being developed meet in relation to the target application?
  - What function is the printed circuit board to be developed intended to perform in the target application?
  - What environmental conditions will the printed circuit board be exposed to, or must withstand? (→ limits of operating temperature range, radiation, vibrations, mechanical shock resistance, electromagnetic interference)

Project Seminar – Basics PCB Design

- ***Example 2nd-Order-LPF***

Second order analog low-pass filter
→ 2-layer design

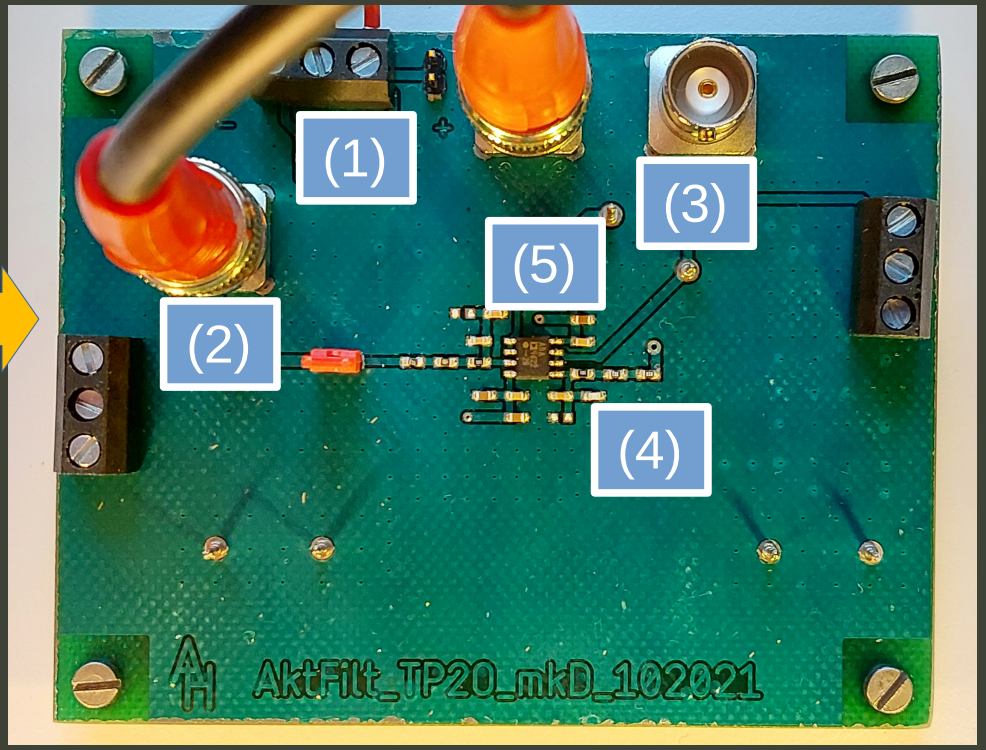
Printed Circuit Board



Project Seminar – Basics PCB Design

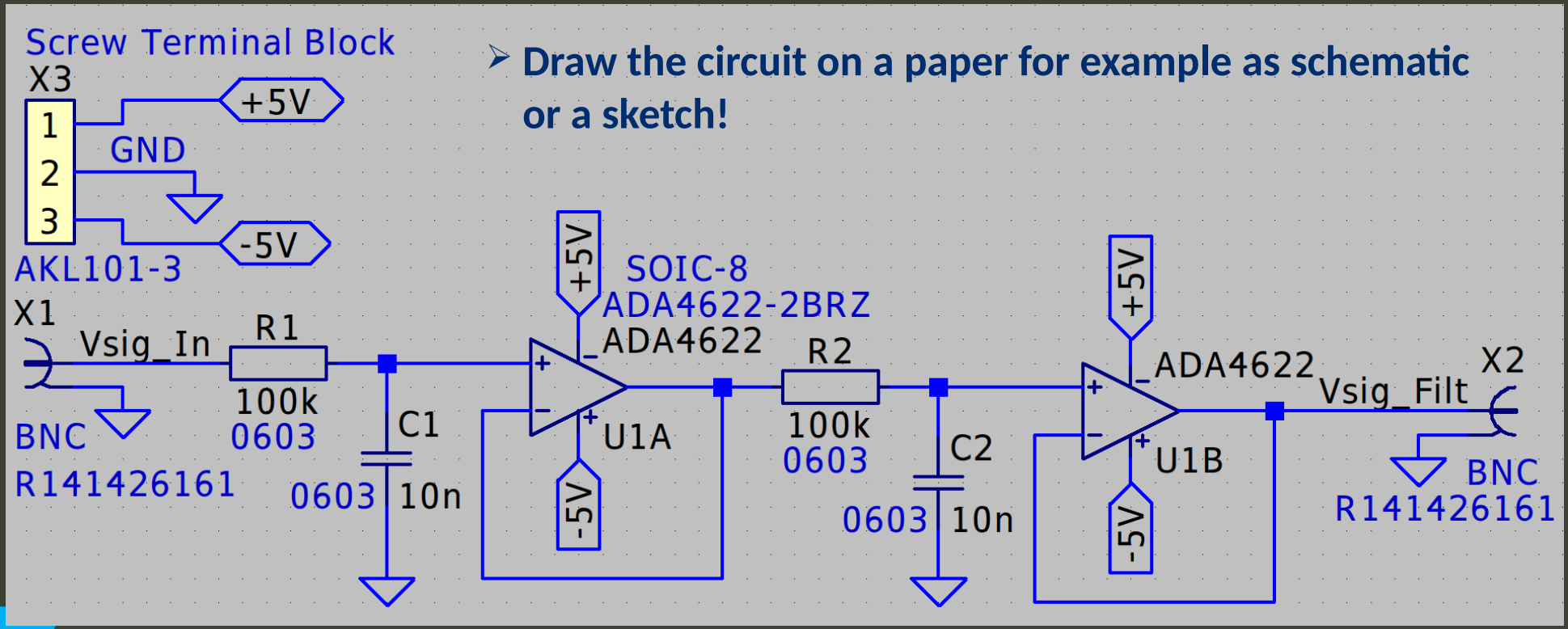
- (1) Screw terminal block → Power supply
- (2) BNC connector → Input
- (3) BNC connector → Output
- (4) capacitors, resistors
- (5) Integrated circuits (IC) → Operational Amplifiers

Printed Circuit Board



Project Seminar – Basics PCB Design

- ***Example 2nd-Order-LPF***

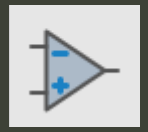


- **Draw the circuit on a paper for example as schematic or a sketch!**

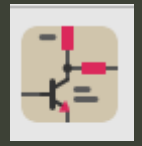
Project Seminar – Basics PCB Design

How are we going to translate that electrical theory into an actual physical design in KiCad?

- First of all make sure that all required component libraries are available to create a clean and good readable schematic.
- If necessary, create the missing component libraries (symbol, footprint, 3d-model).



- Create a clean and good readable schematic in KiCad.



Project Seminar – Basics PCB Design

How are we going to translate that electrical theory into an actual physical design in KiCad?

- Enter the values of all components.

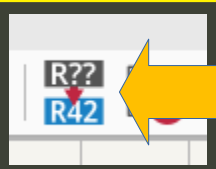
- Assign the corresponding packages to all components.

Package assignment

Project Seminar – Basics PCB Design

How are we going to translate that electrical theory into an actual physical design in KiCad?

- Carry out the annotation of all components.



Annotation

- Fill out the description field of the schematic sheet.
- Carry out a final check of the schematic (Electrical Rule Check, → ERC) !

ERC

KiCad

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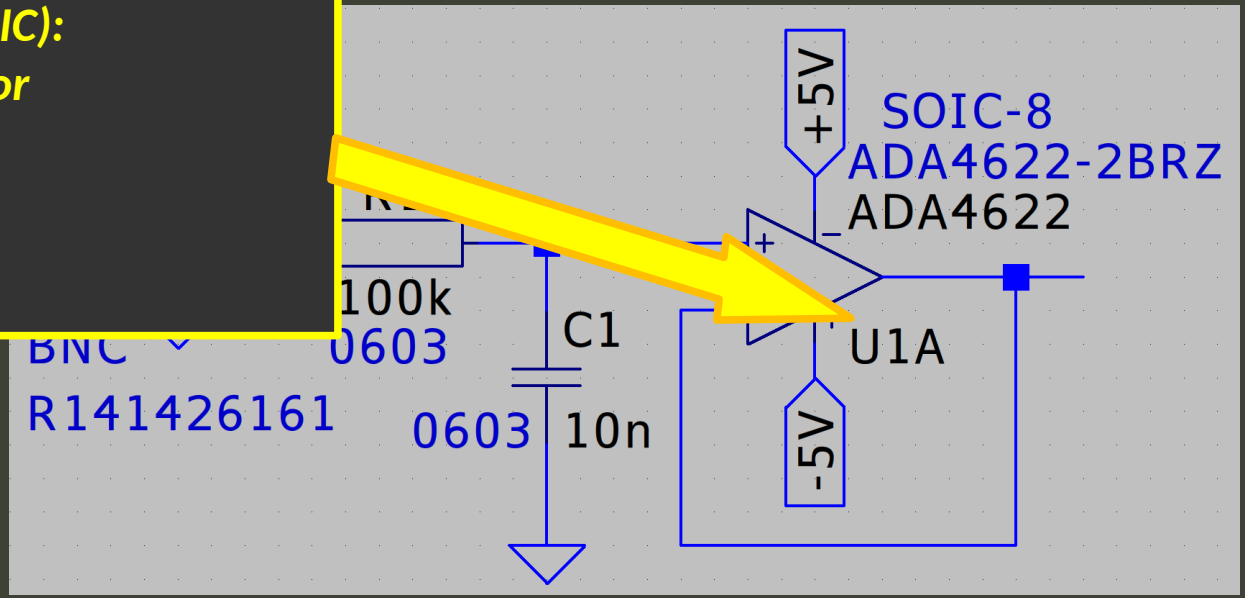
- ***Example:***

Annotation & value of U1 (IC):
- ***U → Reference designator***
- ***1 → id number***
- ***A → sub-circuit***
- ***ADA4622-2 → Value***

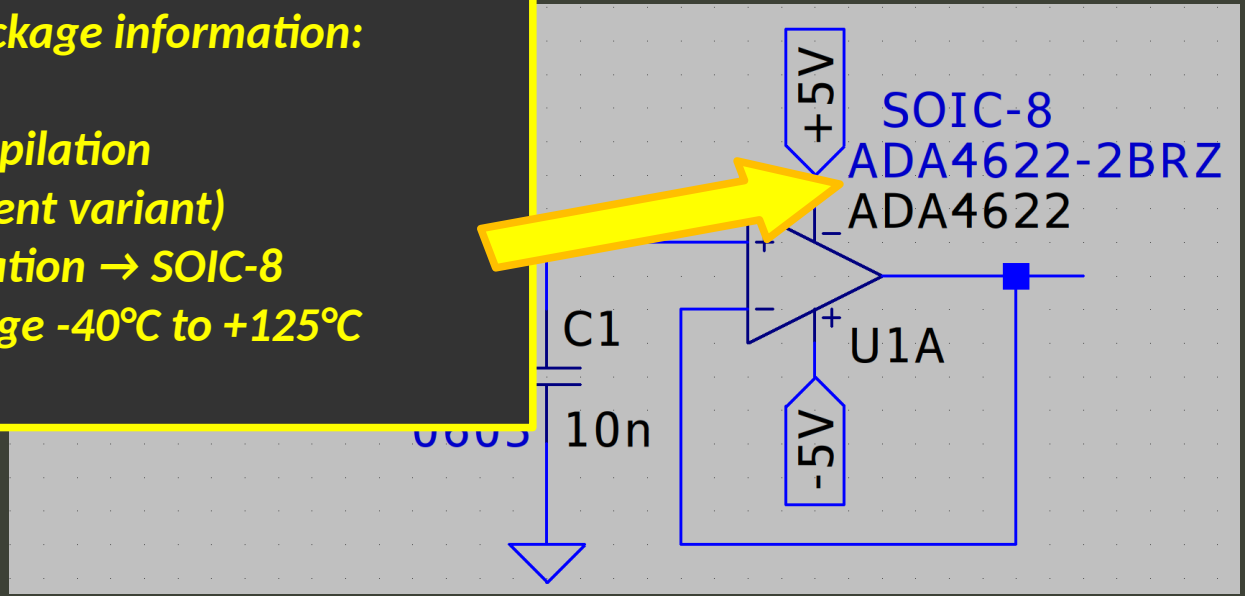


Project Seminar – Basics PCB Design

- ***Example:***

Value extension with package information:

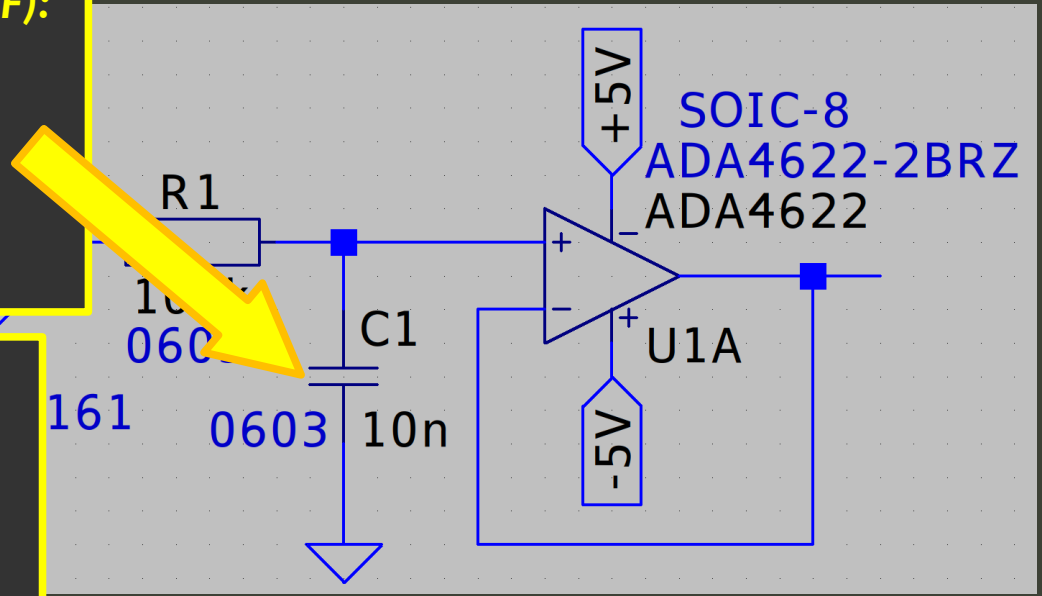
- ***ADA4622-2BRZ***
  → value-package-compilation
- ***B → B-grade (component variant)***
- ***RZ → package information → SOIC-8***
  → temperature range -40°C to +125°C



Project Seminar – Basics PCB Design

- ***Example:***

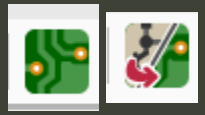
Annotation & value example C1 (100nF):
- ***C → Reference designator***
- ***1 → id number***
- ***10n → capacitance (value) 10nF***

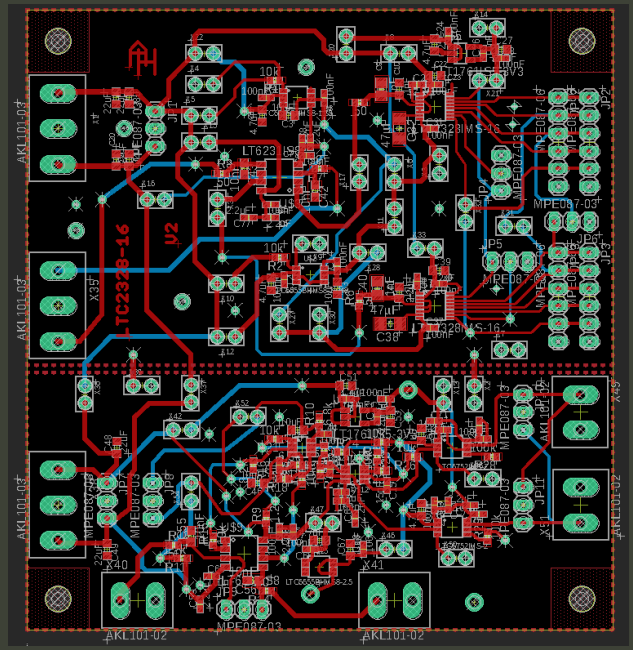


Also possible:
- ***R15A, R15B***
  → resistor network
- ***U1A, U1B***
  → dual operational amplifier

Project Seminar – Basics PCB Design

How are we going to translate that electrical theory into an actual physical design in KiCad?

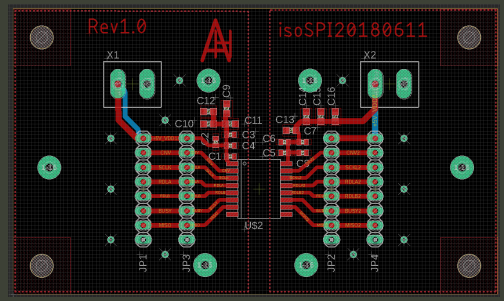
- transfer all components from schematic to the PCB layout editor 
- PCB layout is one of the most exciting and challenging part of PCB design
- sometimes like a giant puzzle, that have to be solved considering technical issues, some original thinking and creativity
  - In addition it is also necessary to think of the manufacturing process of the PCB!



Project Seminar – Basics PCB Design

How are we going to translate that electrical theory into an actual physical design in KiCad?

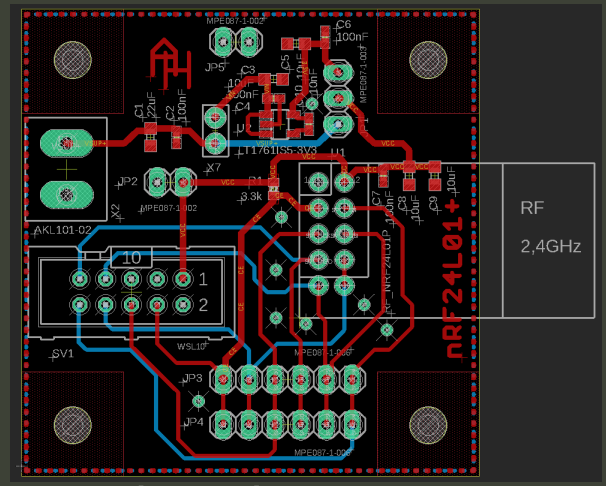
- you are likely designing a PCB layout to have it made physically by a manufacturer
- you have to keep several considerations in the back of your mind while designing PCB
  - many manufacturers offer design rules to avoid problems in the manufacturing process (information about allowed drill-sizes and details of traces for example width and spacing).



Project Seminar – Basics PCB Design

How are we going to translate that electrical theory into an actual physical design?

- you have also to think about the placement and orientation of the components
- do not put your components too close together
- leave enough room between parts to connect them in your routing-process with copper tracks
- make sure that sets of components for example LEDs or resistors facing the same direction
  - this will make it much easier to install, test and inspect your board by the manufacturer



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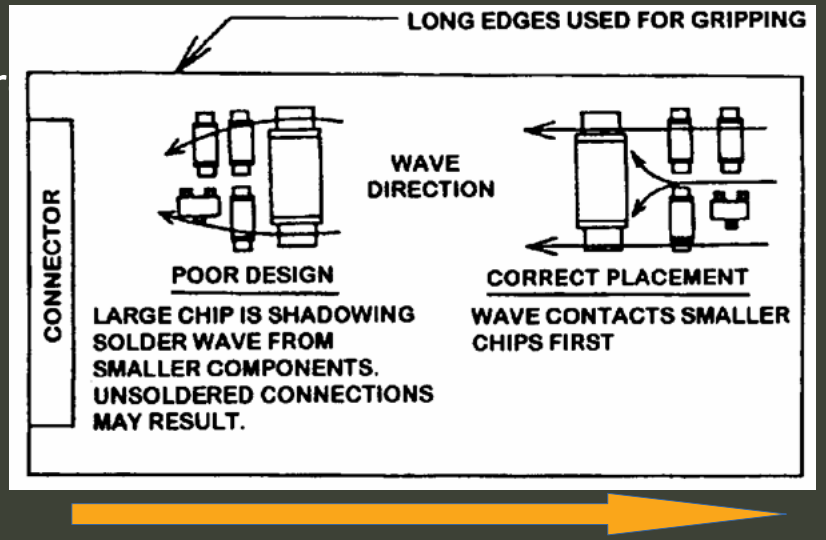
How are we going to translate that electrical theory into an actual physical design?

- board will be send through a soldering oven to connect all of the parts to your bare circuit boar
- make sure that taller components are not blocking smaller ones (causes: poorly connected solder joints)
  - be mindful of their size in the three-dimensional space (!)



direction of movement through soldering wave

(bottom side view)

Project Seminar – Basics PCB Design

- ***Example soldering wave machine:***

ersa POWERFLOW AIR H

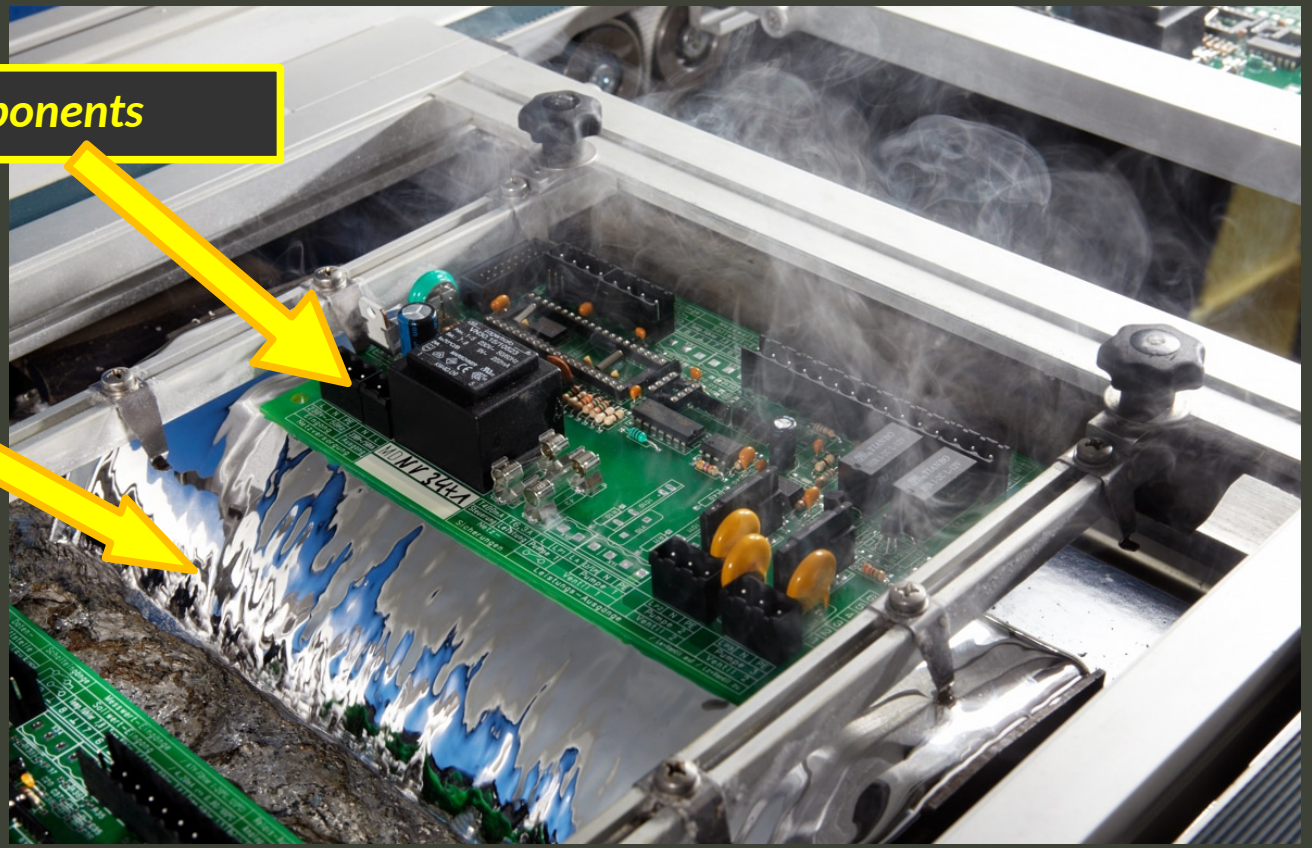
POWER-Wellenlöten mit
flexibler Lötbad-Technologie

Project Seminar – Basics PCB Design

- ***Example soldering wave:***

PCB with through-hole components

Liquid solder



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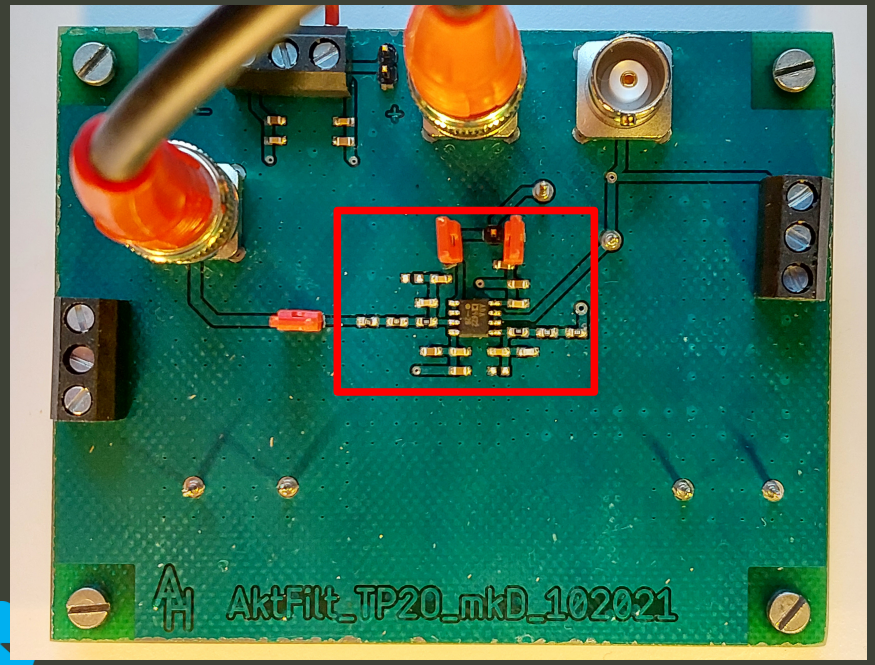
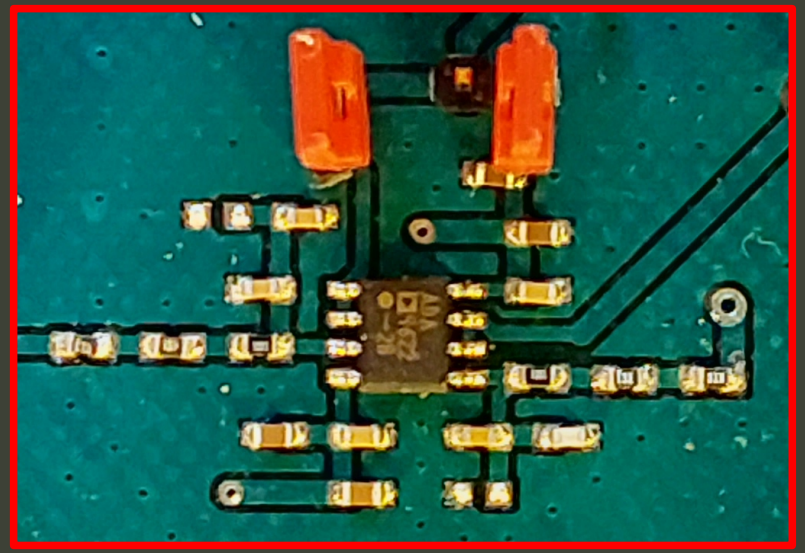
How are we going to translate that electrical theory into an actual physical design?



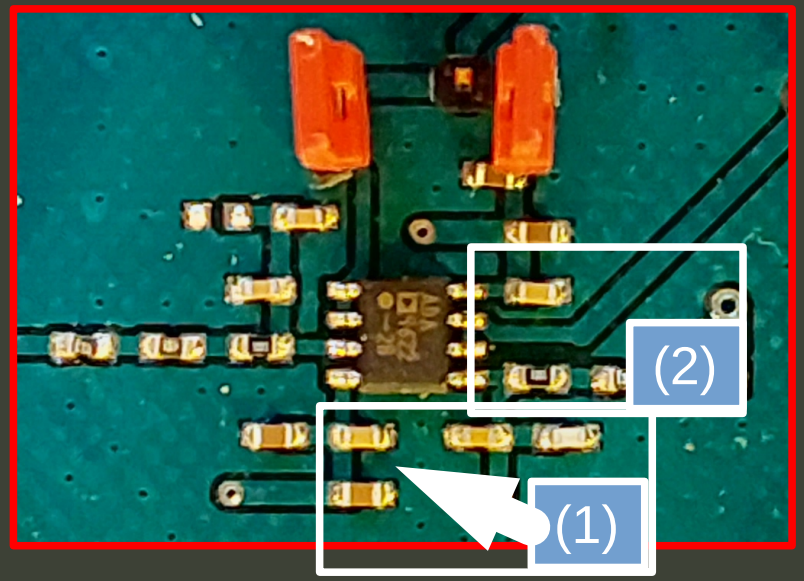
marked area enlarged



Project Seminar – Basics PCB Design

How are we going to translate that electrical theory into an actual physical design?

- (1) place the bulk (buffer) and bypass capacitors as close as possible to the supply pin of every IC
- (2) avoid sharp angles and 90-degree angles between tracks; use 45-degree angles instead



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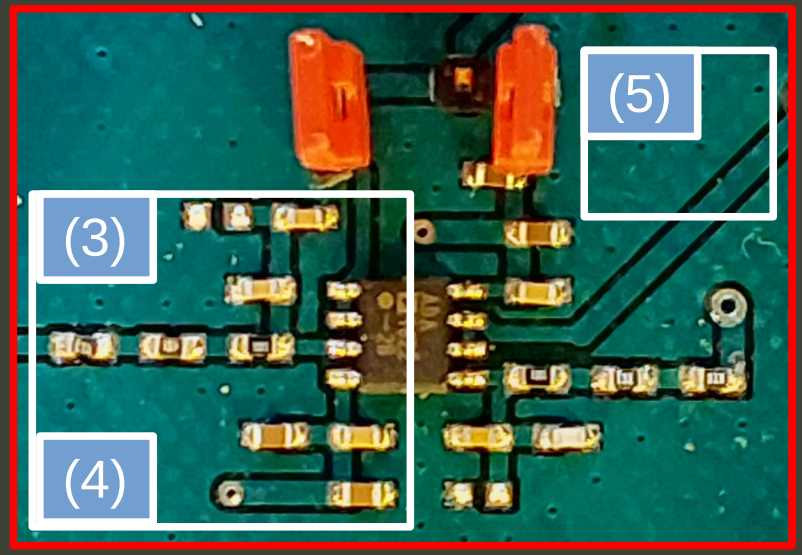
How are we going to translate that electrical theory into an actual physical design?

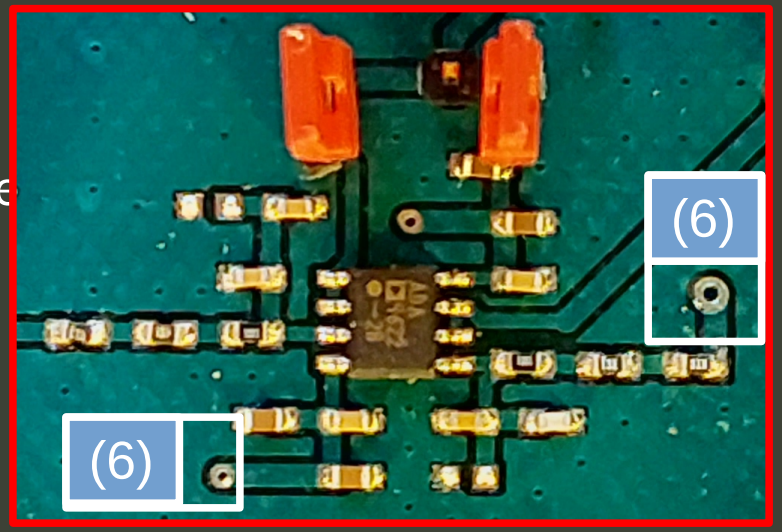
- (3) keep the tracks as short as possible or as absolutely required
- (4) select the conductor track widths so that the required current can be conducted
- (5) select isolation (distance) between the conductor tracks according to the rated voltage & environment



Project Seminar – Basics PCB Design

How are we going to translate that electrical theory into an actual physical design?

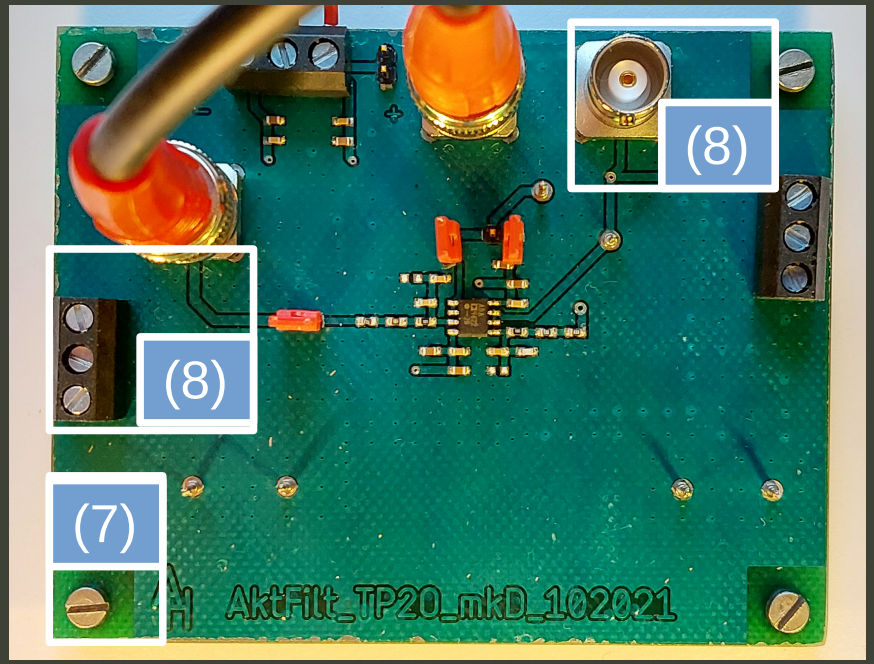
- (6) use vias to connect tracks on different layers
- only use vias where they are absolutely necessary
- pay attention to the permissible impedance of the individual conductor tracks
- do not place analog signal lines close to digital ones !



Project Seminar – Basics PCB Design

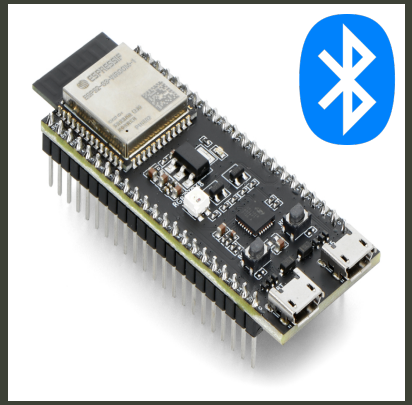
How are we going to translate that electrical theory into an actual physical design?

- (7) think about positioning of mounting holes (installation, handling)
- (8) think about positioning of connectors (installation, handling)
- leave enough space between inductors and other components (coupling, over-voltage, damages)
  - do not forget the final design rule check (DRC)!

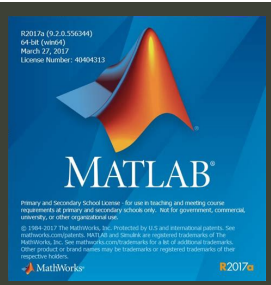


Project Seminar - Project Introduction

- ***Fuel cell monitoring and measurement board:***



GNU Octave



Fuel cell (500W)

Measurement board with ESP32-S3 micro-controller and DAQ-board software

DAQ - Software

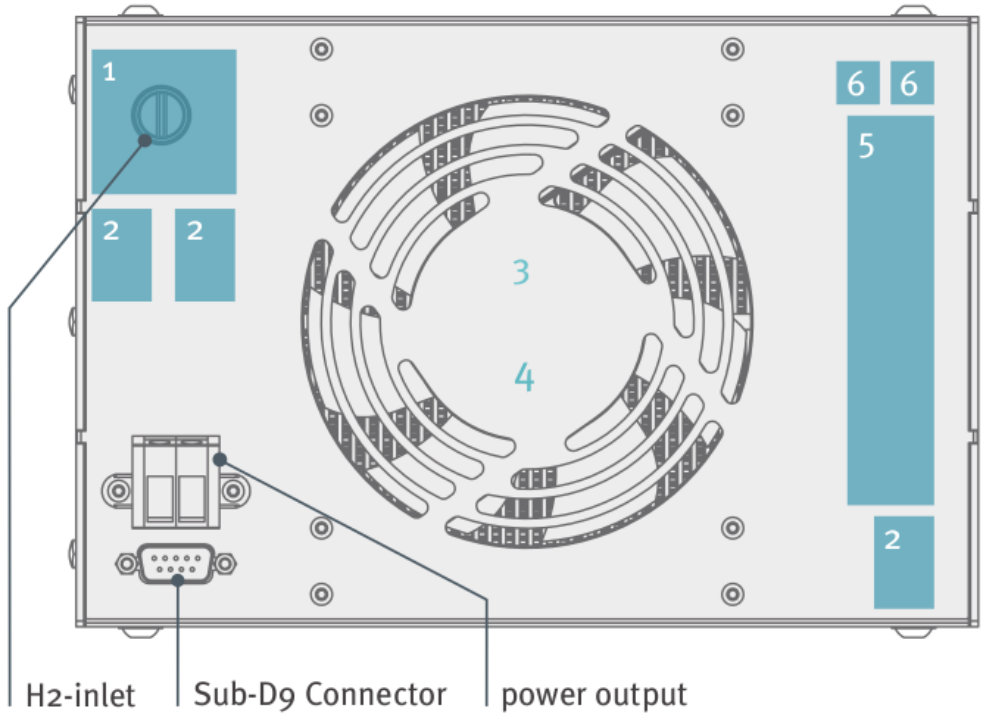
Project Seminar - Project Introduction

- ***Technical data of the fuel cell:***

technology	PEM fuel cell module with integrated peripheral components
operation power	24 - 45 V DC
electrical output	450 W nominal/ 500 W peak
current	18 A at nominal power
efficiency	~ 45 % at 500 W
dimensions (length x width x height)	232 mm x 157 mm x 168 mm
weight	~ 3.9 kg
fuel supply	hydrogen 3.0
pressure range (fuel)	up to 10 bar (internal reduction)
integrated safety	H2 pressure & temperature sensor
H2-inlet	G 1/8“ inside thread
electr. power connector	4 mm² screw terminal
controller interface	SUB-D9 (RS232)

Project Seminar - Project Introduction

- ***Integration of basic peripheral components of the fuel cell:***



Hydrogen supply:

1 pressure regulator

2 shut off/safety valves

Air supply/Cooling:

3 blower/fan

4 air filter

Electronics:

5 control circuits

6 sensors

Project Seminar - Project Introduction

- ***Structure and function of a PEM fuel cell (on cell level):***

Hydrogen input

BPP: Bipolar plate

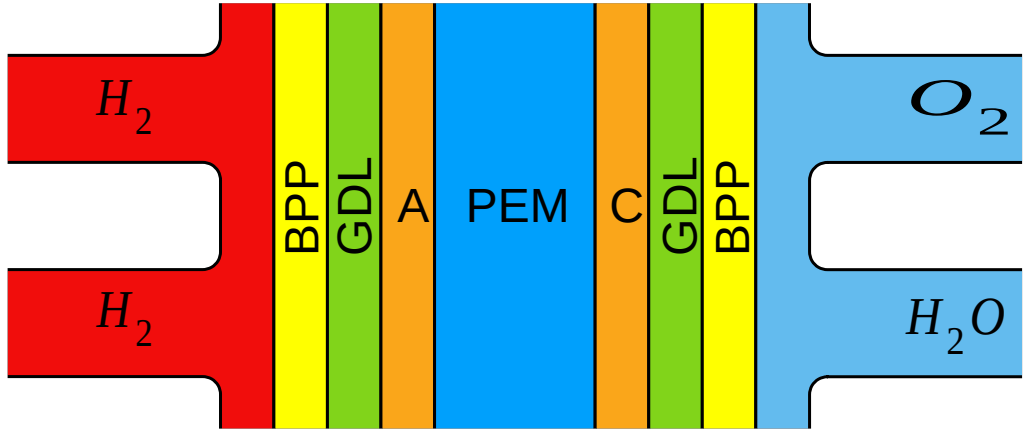
GDL: Gas diffusion layer

A: Anode

PEM: Proton Exchange Membran

C: Cathode

Oxygen input & Water outlet



Project Seminar - Project Introduction

- ***Structure of a PEM fuel cell (on cell level):***

BPP: Bipolar plate

  - In a multi-cell or stack configuration, the primary function of the bipolar plate is to provide a physically and electrically conductive connection between the anode of one cell and the cathode of the adjacent cell.
  - In addition, the bipolar plate is responsible for directing the reaction gases into the reaction zone. To this end, flow fields are milled or pressed into both sides of the plates, through which hydrogen flows on one side and air is supplied on the other.
  - A bipolar plate consists of the two poles (hence the name bipolar) of a single fuel cell: the hydrogen-carrying anode plate (the negative (-) pole) and the cathode plate (the positive (+) pole) for supplying the reaction air. The plates also regulate the removal of water vapor and the release of thermal and electrical energy.

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- ***Structure of a PEM fuel cell (on cell level):***

GDL: Gas diffusion layer

  - The gas diffusion layer ensures that hydrogen and oxygen are evenly distributed to the electrodes, which is necessary for the electrochemical reaction.
  - The even distribution makes the electrochemical reaction more efficient, which increases the fuel cell's performance.
  - The gas diffusion layer removes the water from the chemical reaction

Project Seminar - Project Introduction

- ***Structure and fucntion of a PEM fuel cell (on cell level):***
  - Oxidation of the hydrogen gas at the anode (gas conductive & catalytic electrode)

$$H_2 \rightarrow 2H^+ + 2e^-$$

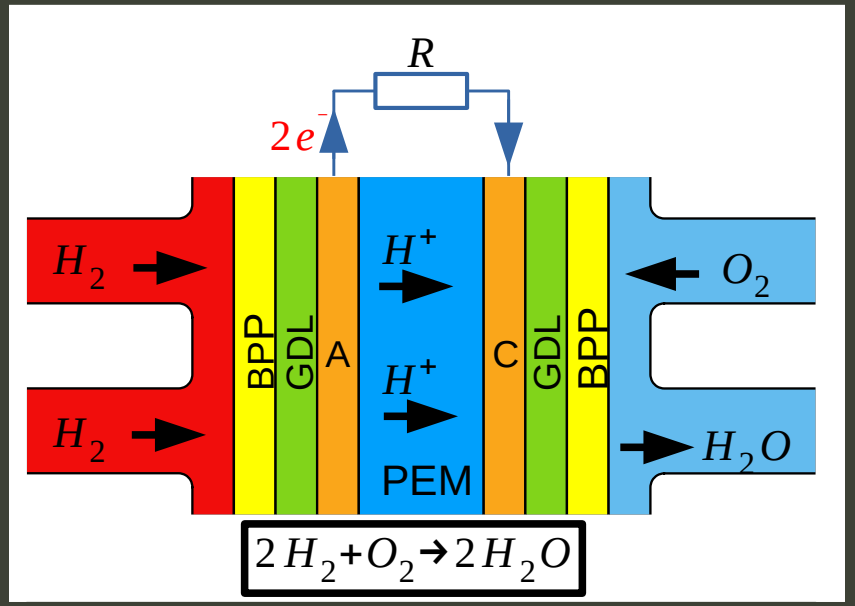
  - Reduction of the oxygen gas at the cathode (gas conductive & catalytic electrode)

$$O_2 + 4e^- \rightarrow 2O^{2-}$$

  - The protons migrate through the proton exchange membrane (PEM) to the reduced oxygen.

$$2O^{2-} + 4H^+ \rightarrow 2H_2O$$

Anode | Cathode



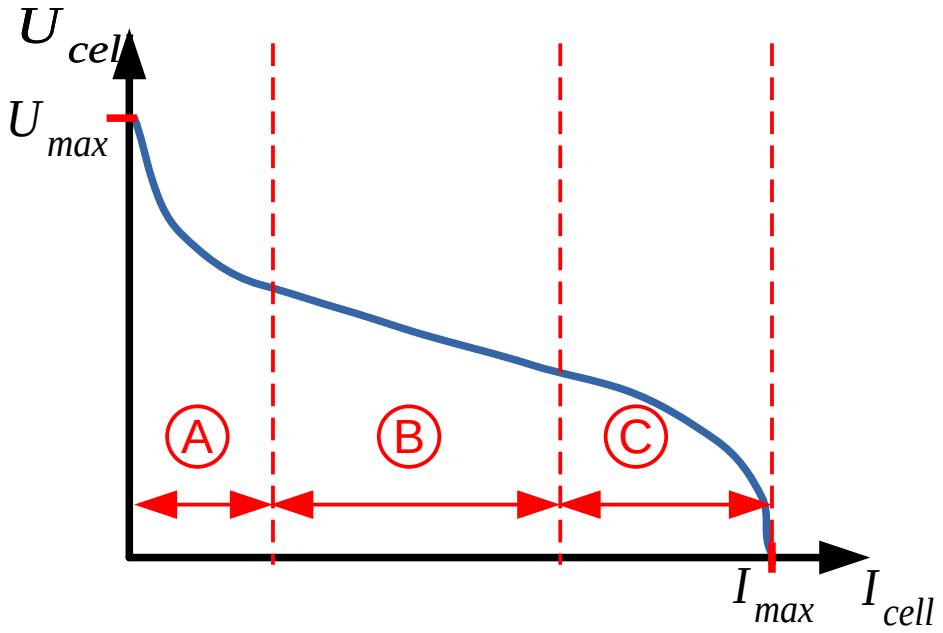
Project Seminar - Project Introduction

- ***Characteristic curve of a fuel cell:***

Ⓐ Electro-kinetic area

Ⓑ Ohmic area

Ⓒ Diffusion-controlled area



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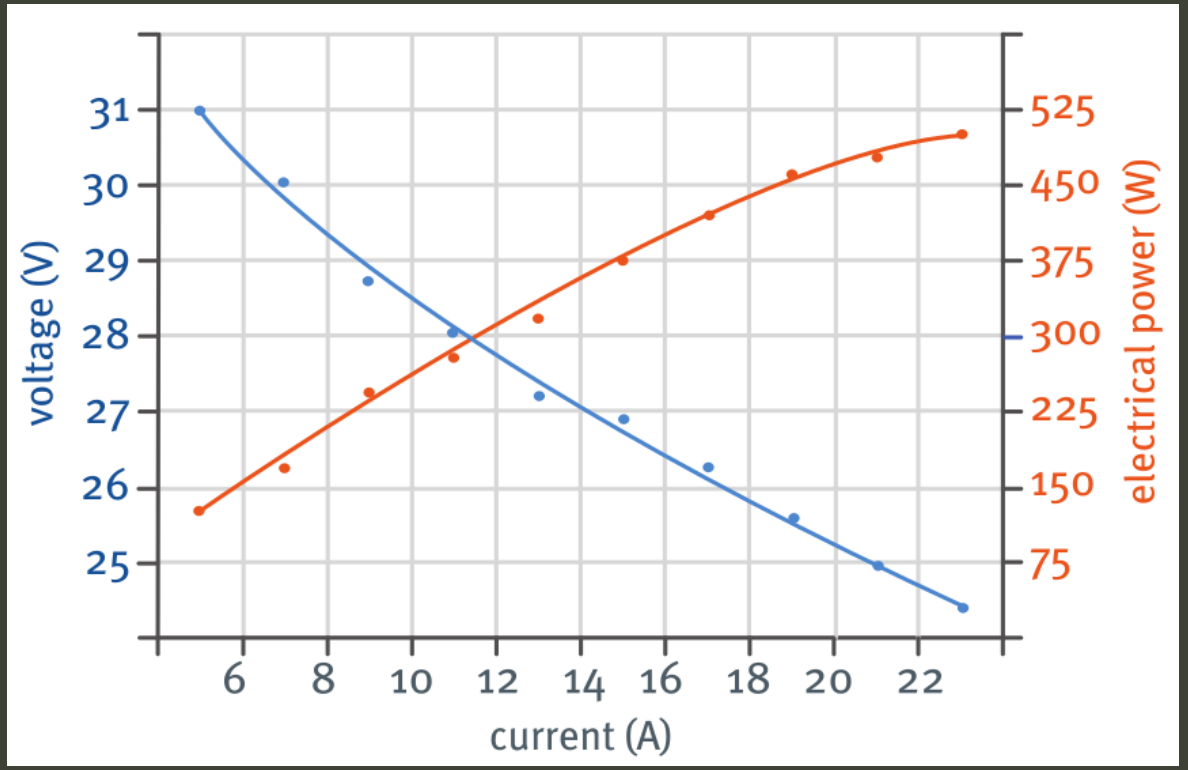
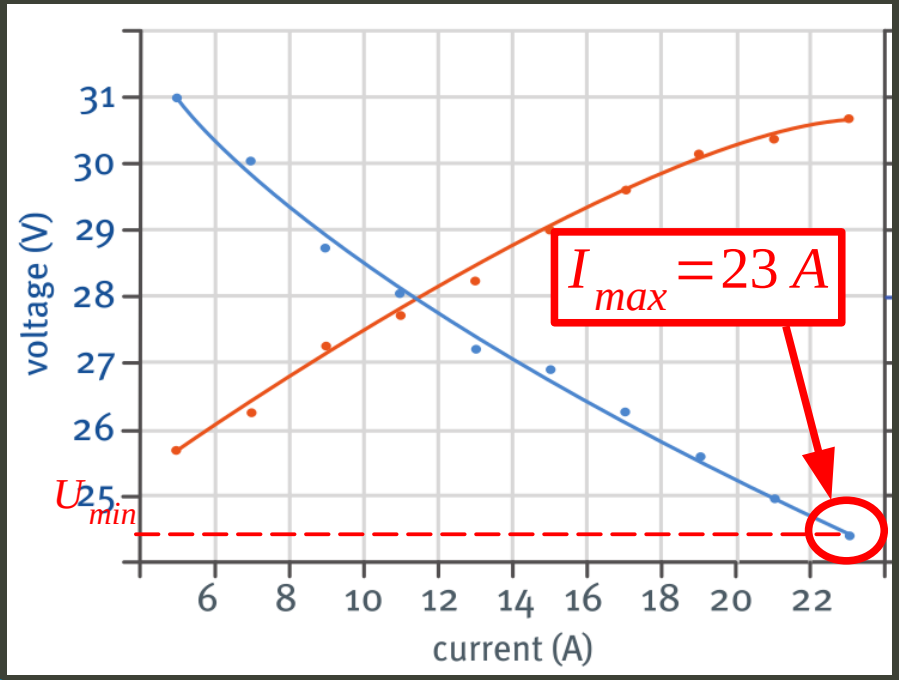
- ***Characteristic curve of the fuel cell in the project:***



Project Seminar - Project Introduction

- ***Characteristic of the fuel cell in the project:***



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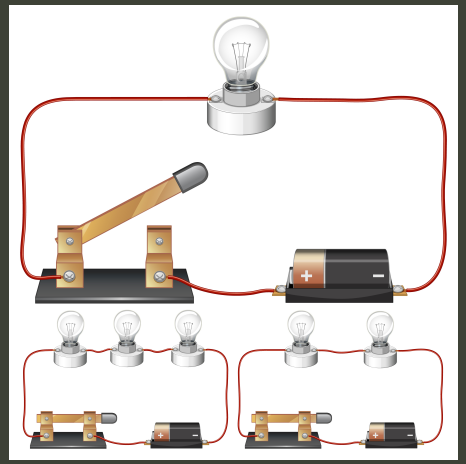
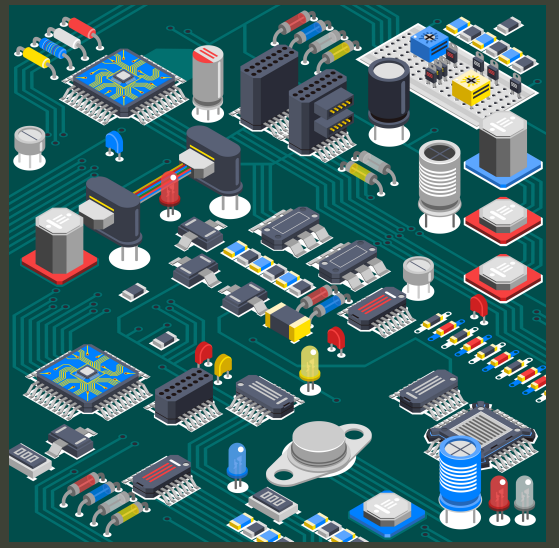
dimensions (length x width x height)	232 mm x 157 mm x 168 mm
weight	~ 3.9 kg
fuel supply	hydrogen 3.0
pressure range (fuel)	up to 10 bar (internal reduction)
integrated safety	H2 pressure & temperature sensor
H2-inlet	G 1/8“ inside thread
electr. power connector	4 mm² screw terminal
controller interface	SUB-D9 (RS232)

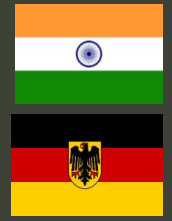
Project Seminar - Project Introduction

Let's talk about analog circuits ...





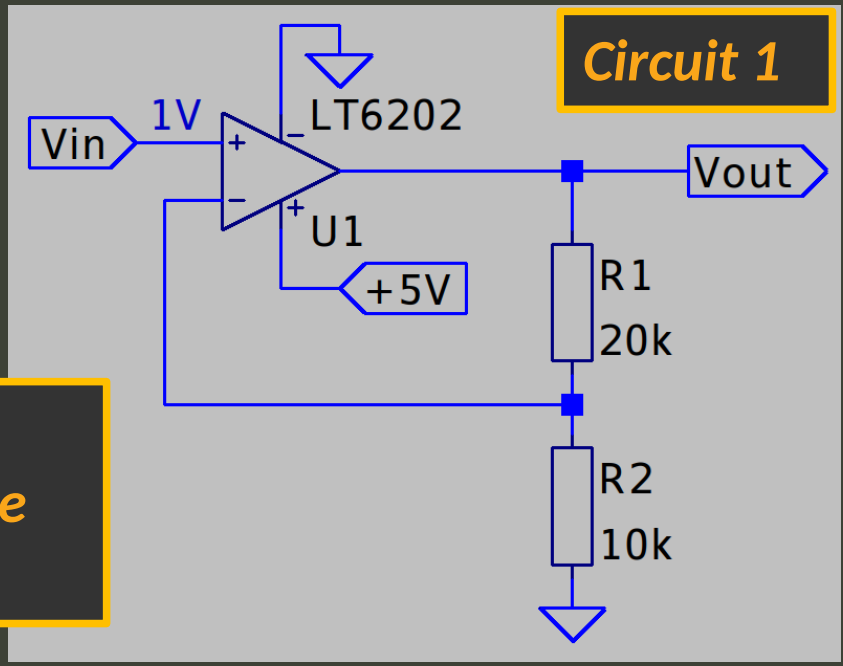
Project Seminar - Project Introduction



- ***Analog circuits → non-inverting operational amplifier:***
  - LT6202 as non-inverting operational amplifier

→ Task: What are the characteristics of operational amplifiers? Figure it out!

→ Task: How can you calculate the output voltage Vout? Figure it out!



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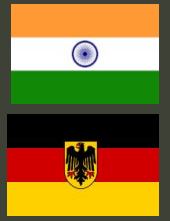
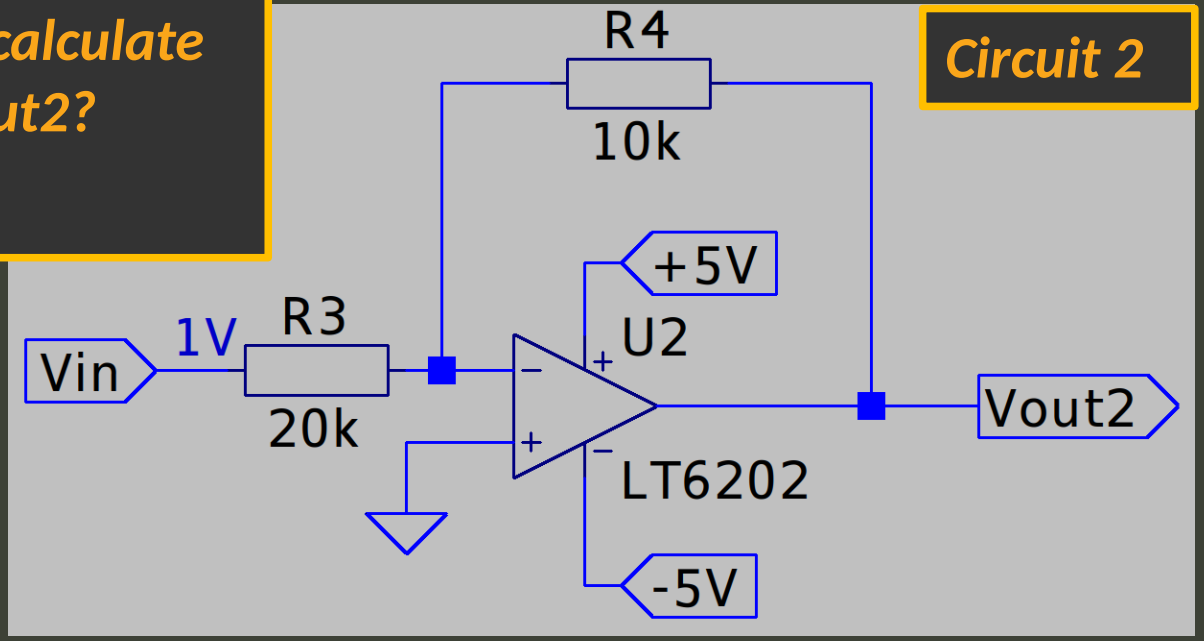


- ***Analog circuits → inverting operational amplifier:***
  - LT6202 as inverting operational amplifier

→ Task: How can you calculate the output voltage Vout2? Figure it out!



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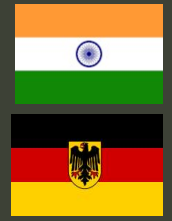
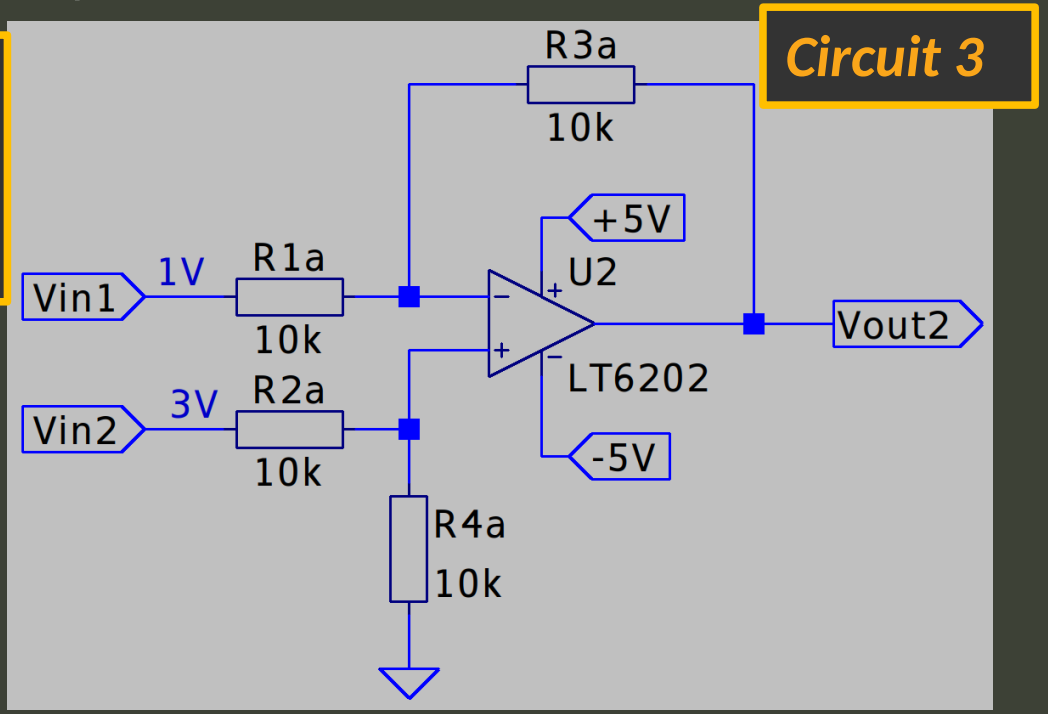


- ***Analog circuits → Differential operational amplifier:***
  - LT6202 as differential operational amplifier

→ Task: How can you calculate the output voltage Vout2? Figure it out!



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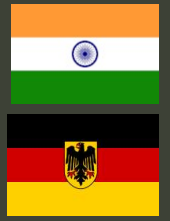
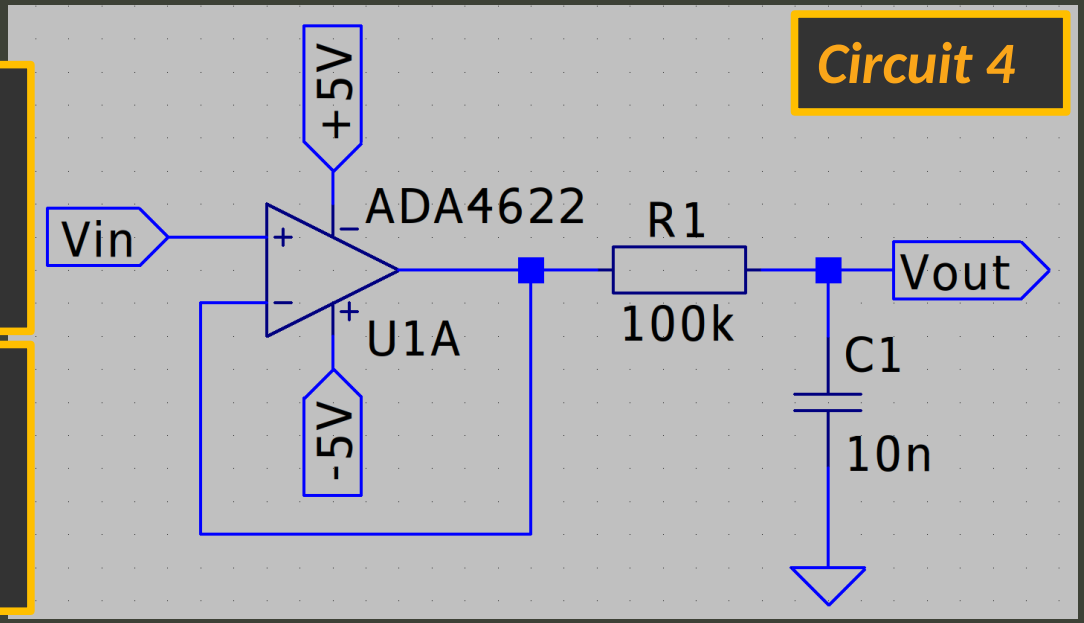


- ***Analog circuits → Impedance amplifier → Voltage follower:***
  - ADA4622 as impedance amplifier (voltage follower) with analog low-pass filter

→ Task: How can you calculate the dc output voltage Vout? Figure it out!

→ Task: How can you calculate the ac output voltage Vout? Figure it out!



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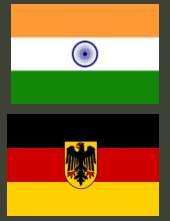

- ***Analog circuits → exercises:***
  - Exercise 1: List the characteristics of an ideal operational amplifier!
  - Exercise 2: Calculate the output voltage Vout in Circuit 1!
  - Exercise 3: Calculate the output voltage Vout2 in Circuit 2!
  - Exercise 4: Calculate the output voltage Vout2 in Circuit 3!
  - Exercise 5: Calculate the output voltage Vout in Circuit 4 in dB!
    Case 1: Vin is a DC-voltage of 1V
    Case 2: Vin is a AC-voltage u(t)= 1V*sin(2*π*200Hz*t)

→ TASK: Answer these questions in your report!

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- ***Analog circuits → exercises:***
  - Exercise 6: What is a reference voltage source ? For which applications are reference voltage sources required ? Give examples!
  - Exercise 7: What is a SAR ADC ? For which applications are SAR ADCs required and how does it work in detail?

→ TASK: Answer these questions in your report!

Project Seminar - Schedule

- ***Lessons and tasks:***
  - independent study – repetition & solving exercises
  - until 19.10.2026: Write chapter 3 of your project report:
    → Title: “Basics in PCB-Design”
    → Write a summary about what you know about PCB-Design ?
  - until 19.010.2026: Write chapter 4 of your project report:
    → Title: “PEM-Fuel-Cell”
    → answer: What do you know about PEM fuel cells? What are the basic peripheral components of the fuel cell? How does the fuel cell work on cell level? Draw and explain the characteristic curve of a PEM fuel cell!
  - Lesson 4 – Working on project

… schedule is continously updated!

Project Seminar - WS2026/27

Thank you !
Have a nice day!